

The Shift in Disguise: A Non-Faithful Functor from Fields to Symbolic Dynamics

Abstract

One element of a field becomes one sequence.

The Frobenius map becomes the shift.

Coprimality becomes a minimal rotation.

The functor is real, but it forgets.

We say exactly what it keeps, and what it drops.

Everything is checked in Lean 4 / Mathlib.

Conference paper call — short, slow-read submission.

1 The picture

Take a finite field $\mathbb{F}_{2^n}$.

Pick one element v .

Square it, again and again:

$$v, \quad v^2, \quad v^4, \quad \dots, \quad v^{2^{n-1}}.$$

That is a sequence.

It repeats with period dividing n .

One field element. One periodic sequence.

2 The shift

Squaring once moves the whole sequence one step left:

$$(v, v^2, v^4, \dots) \xrightarrow{x \mapsto x^2} (v^2, v^4, v^8, \dots).$$

That is the *shift* σ .

So the Frobenius map **is** the shift.

$$x \mapsto x^2 \quad \longleftrightarrow \quad \sigma.$$

And the k -th power of Frobenius is the k -fold shift.

$$x \mapsto x^{2^k} \quad \longleftrightarrow \quad \sigma^k.$$

Algebra on the left. Dynamics on the right.

3 The encoding, in Lean

The orbit sequence:

```
def frobeniusOrbitSeq (p : ℕ) (x : F) : ℕ → F :=  
  fun k => x ^ (p ^ k)
```

One step is one Frobenius:

```
theorem frobeniusOrbitSeq_succ (p : ℕ) (x : F) (k : ℕ) :  
  frobeniusOrbitSeq p x (k + 1)  
    = (frobeniusOrbitSeq p x k) ^ p
```

The recurrence *is* the shift rule.

4 Cool fact 1 — counting fixed points

How many sequences does σ^k fix?

How many v satisfy $v^{2^k} = v$?

Same question. Same answer.

$$\#\{v : v^{2^k} = v\} = 2^{\gcd(k, n)}.$$

The shift side counts *necklaces*:

$$\#\text{Fix}(\sigma^k) = 2^{\gcd(k, n)}.$$

One formula, read two ways.

5 Cool fact 1, in Lean

The field side:

```
theorem frobenius_fixed_card (n k : ℕ) (hn : 0 < n) (hk : 0 < k)
  (hcard : Fintype.card F = 2 ^ n) :
  Nat.card {x : F | x ^ (2 ^ k) = x} = 2 ^ Nat.gcd k n
```

The shift side:

```
theorem necklace_orbit_count (n d : ℕ) [NeZero n] (hn : 0 < n) :
  Fintype.card {w : ZMod n → ZMod 2 | ∀
    i, w i = w (i + (d : ZMod n))}
  = 2 ^ Nat.gcd d n
```

Same right-hand side. No coincidence.

6 Cool fact 2 — minimal rotations

Now ask: when is σ^k a *minimal* rotation of $\mathbb{Z}/n\mathbb{Z}$?

Exactly when

$$\gcd(k, n) = 1.$$

A minimal rotation sweeps the whole cycle.

So it can fix a sequence **only if the sequence is constant**.

Constant sequence $\iff$ the element lives in $\mathbb{F}_2 = \{0, 1\}$.

No invariant sub-coordinate. Nowhere to hide.

7 Cool fact 2, in Lean

Theorem 1. *If $\gcd(k, n) = 1$, then*

$$v^{2^k} = v \iff v \in \{0, 1\}.$$

```
lemma frob_fixed_iff_GF2 (hn : Fintype.card F = 2 ^ n)
  (hcop : Nat.Coprime k n) :
  v ^ (2 ^ k) = v ↔ (v = 0 ∨ v = 1)
```

Coprimality = minimality. The dynamical reading of a number-theory side condition.

8 The big idea

The field proof of coprimality facts splits by the parity of n .

The dynamical reading does not.

“A minimal rotation fixes only constant sequences.”

One sentence. Every n . No parity.

That uniformity is the payoff of the encoding.

Move the question to where it is easy. Then read the answer back.

9 The functor

This is not one example. It is a *map of theories*.

$$\mathcal{F} : \underbrace{\text{Frobenius orbits on } \mathbb{F}_{2^n}}_{\text{algebra}} \longrightarrow \underbrace{\text{shift orbits on } \{0,1\}^{\mathbb{Z}/n\mathbb{Z}}}_{\text{dynamics}}$$

At the level of orbit theories the two sides agree.

```
theorem dynamics_algebra_bridge (n : ℕ) :
  MoritaEquiv (frobeniusOrbitTheory 2 n) (shiftOrbitTheory n)
```

So invariants cross for free:

```
theorem bridge_transfer (n : ℕ) (I : MoritaInvariant)
  (h : I.prop (frobeniusOrbitTheory 2 n)) :
  I.prop (shiftOrbitTheory n)
```

10 But the functor forgets

The shift sees one operation: the **multiplicative** Frobenius.

It never sees the field **addition**.

$$\sigma(x) = x^2 \quad \text{respects } \times, \quad \text{but } + \text{ is invisible to } \sigma.$$

So the functor is **non-faithful**:

different additive phenomena collapse to the *same* shift dynamics.

A conjugacy of the $\times$ -structure is not a conjugacy of the field.

11 Why this matters

APN is an **additive** property:

$$f(x+a) - f(x) \text{ is at most two-to-one.}$$

The shift forgot $+$.

So a *naïve* topological conjugacy with the shift **cannot** compute the deep APN facts.

The $+/\times$ clash breaks the dream of “just work on the model.”

The bridge is honest about its own limit.

12 What survives: the spectral layer

Replace the conjugacy by the *transfer operator* — the character sum.

$$W_f(a, b) = \sum_x (-1)^{\mathrm{Tr}(ax + bf(x))}.$$

partition function	Walsh sum W_f
pressure / moments	$\sum W^2, \sum W^4$
expansion rate	2-adic valuation of W
fundamental domain	Frobenius orbit (cyclotomic coset)

Spectral data crosses the $+/\times$ clash. A bare conjugacy does not.

13 APN, as a dynamical fibre condition

Even without conjugacy, APN *is* a statement about fibres of a dynamics.

$$\text{APN}(x \mapsto x^d) \iff \text{every fibre of } x \mapsto (x + a)^d + x^d \text{ has size } \leq 2.$$

```
theorem apn_small_fibers (d : ℕ) :  
  IsAPN (powerFunction d : F → F) ↔ ∀  
    a : F, a ≠ 0 → ∀ b : F,  
      Fintype.card { x // differenceDynamics d a x = b } ≤ 2
```

The dynamical picture frames the question — the spectral layer answers it.

14 What we claim

- One element $\mapsto$ one periodic sequence.
- Frobenius = shift; $x^{2^k} = \sigma^k$.
- $\#\text{Fix} = 2^{\gcd(k,n)}$, both sides.
- Coprime = minimal rotation = fixes only constants.
- A genuine functor (Morita bridge of orbit theories).
- **Non-faithful**: it keeps $\times$, forgets $+$.
- So deep APN needs the spectral layer, not a bare conjugacy.
- All boxed statements are machine-checked.

Read slowly. The shift was there all along.